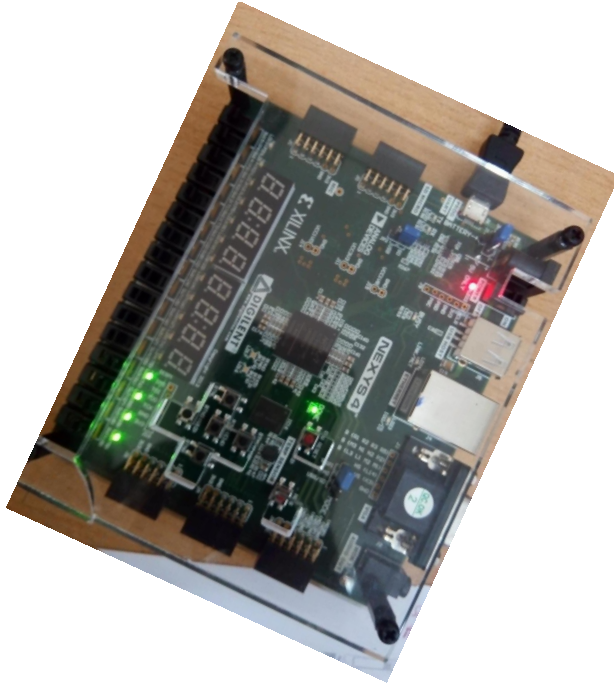




REALISE PAR :

Limème Hend
ouerghemmi Sarra

classe

2 gec 1-1

TP1

Additionneur

1) code vhd de demi additionneur

ModelSim ALTERA STARTER EDITION 6.4a

File Edit View Compile Simulate Add Source Tools Layout Window Help

Workspace

Instance	Design unit	Design unit type	Visibility	States	State hits	State misses	State %
demi_add	demi_add(a...	Architecture	+acc=v				
line_10	demi_add(a...	Process	+acc=v				
line_11	demi_add(a...	Process	+acc=v				
standard	standard	Package	+acc=v				
std_logic_1164	std_logic_1...	Package	+acc=v				
std_logic_arith	std_logic_ar...	Package	+acc=v				
std_logic_unsigned	std_logic_un...	Package	+acc=v				

C:/aaaaa/work/ex.vhd

```
1 library IEEE;
2 use IEEE.STD_LOGIC_1164.all;
3 use IEEE.STD_LOGIC_ARITH.ALL;
4 use IEEE.STD_LOGIC_UNSIGNED.all;
5 entity demi_add is
6     port(a,b: in std_logic; s,cout: out std_logic);
7 end;
8 architecture arch1 of demi_add is
9 begin
10     s <= a xor b ;
11     cout <= (a and b) ;
12 end arch1;
13
```

demadd - [C:/aaaaa/work/demadd/planAhead_run_2/demadd.ppr] - PlanAhead 14.6

File Edit Tools Window Layout View Help

Elaborated Design 3

RTL Netlist

- demi_add
- Nets (4)
- Primitives (2)

I/O Port Properties

Name: b
Direction: Input
Net: b

General Attributes Configure

Properties Clock Regions

Package x Device x RTL Schematic x

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18

A B C D E F G H I J K L M N O P Q R S T U V

I/O Ports

Name	Direction	Neg Diff Pair	Site	Fixed	Bank	I/O Std	Vcco	Vref	Drive Stre...	Slew Type	Pull Type	Off-Chip T...	IN_TERM
All ports (4)													
Scalar ports (4)													
a	Input					LVC MOS 18	1.800				NONE	NONE	
b	Input					LVC MOS 18	1.800				NONE	NONE	
cout	Output					LVC MOS 18	1.800		12 SLOW		NONE	FP_VTT_50	
s	Output					LVC MOS 18	1.800		12 SLOW		NONE	FP_VTT_50	

Td Console Package Pins I/O Ports

FR 11:41 29/01/2019

ISE Project Navigator (P.68d) - C:\vhdlams\tpams\tpams.xise - [demi_add.ucf]

File Edit View Project Source Process Tools Window Layout Help

Design

View: Implementation Simulation

Hierarchy

- tpams
 - xc7a100t-3csg324
 - demi_add - arch1 (demi_add.vh)

No Processes Running

Processes: demi_add - arch1

- Design Summary/Reports
- Design Utilities
- User Constraints
- Create Timing Constraints
- I/O Pin Planning (PlanAhead...)
- I/O Pin Planning (PlanAhead...)
- Floorplan Area/IO/Logic (Pla...)
- Synthesize - XST
- Implement Design
- Generate Programming File
- Configure Target Device
- Analyze Design Using ChipScope

```

1 NET "a" LOC=U9 ;
2 NET "b" LOC=U8 ;
3 NET "cout" LOC=T8 ;
4 NET "s" LOC=R8 ;
5
6 NET "a" IOSTANDARD = LVCMOS18;
7 NET "b" IOSTANDARD = LVCMOS18;
8 NET "cout" IOSTANDARD = LVCMOS18;
9 NET "s" IOSTANDARD = LVCMOS18;
10

```

Design Summary

demi_add.ucf

Console

For further information on the status of this process, see the "_impact.log" file.

Process "Configure Target Device" launched successfully

Ln 1 Col 1 UCF

13:17 27/01/2019

ISE Project Navigator (P.68d) - C:\vhdlams\tpams\tpams.xise - [Design Summary]

File Edit View Project Source Process Tools Window Layout Help

Design

View: Implementation Simulation

Hierarchy

- tpams
 - xc7a100t-3csg324
 - demi_add - arch1 (demi_add.vh)

No Processes Running

Processes: demi_add - arch1

- Design Summary/Reports
- Design Utilities
- User Constraints
- Create Timing Constraints
- I/O Pin Planning (PlanAhead...)
- I/O Pin Planning (PlanAhead...)
- Floorplan Area/IO/Logic (Pla...)
- Synthesize - XST
- Implement Design
- Generate Programming File
- Configure Target Device
- Analyze Design Using ChipScope

Design Overview

- Summary
 - IOB Properties
 - Module Level Utilization
 - Timing Constraints
 - Pinout Report
 - Clock Report
 - Static Timing
- Errors and Warnings
 - Parser Messages
 - Synthesis Messages
 - Translation Messages
 - Map Messages
 - Place and Route Messages
 - Timing Messages
 - Bitgen Messages
 - All Implementation Messages
- Detailed Reports
 - Synthesis Report
 - Translation Report
 - Map Report
 - Place and Route Report
 - Post-PAR Static Timing Report
 - Power Report
 - Bitgen Report
- Secondary Reports

Design Properties

- Enable Message Filtering
- Optional Design Summary Contents
 - Show Clock Report
 - Show Failing Constraints
 - Show Warnings
 - Show Errors

demi_add Project Status

Project File:	tpams.xise	Parser Errors:	No Errors
Module Name:	demi_add	Implementation State:	New
Target Device:	xc7a100t-3csg324	Errors:	
Product Version:	ISE 14.6	Warnings:	
Design Goal:	Balanced	Routing Results:	
Design Strategy:	Xilinx Default (unlocked)	Timing Constraints:	
Environment:		Final Timing Score:	

Detailed Reports

Report Name	Status	Generated	Errors	Warnings	Infos
Synthesis Report					
Translation Report					
Map Report					
Place and Route Report					
Power Report					
Post-PAR Static Timing Report					
Bitgen Report					

Secondary Reports

Report Name	Status	Generated

Date Generated: 01/27/2019 - 12:52:11

Design Summary

Console

For further information on the status of this process, see the "_impact.log" file.

Process "Configure Target Device" launched successfully

Console Errors Warnings Find in Files Results

13:16 27/01/2019

ISE iMPACT (P.68d) - [Boundary Scan]

File Edit View Operations Output Debug Window Help

IMPACT Flows

- Boundary Scan
- SystemACE
- Create PROM File (PROM File Format...
- WebTalk Data

IMPACT Processes

Available Operations are:

- Get Device ID
- Get Device Signature/Usercode
- Read Device Status

TDI

xc7a100t

bypass

TDO

Identify Succeeded

Boundary Scan

done.
PROGRESS_END - End Operation.
Elapsed time = 0 sec.

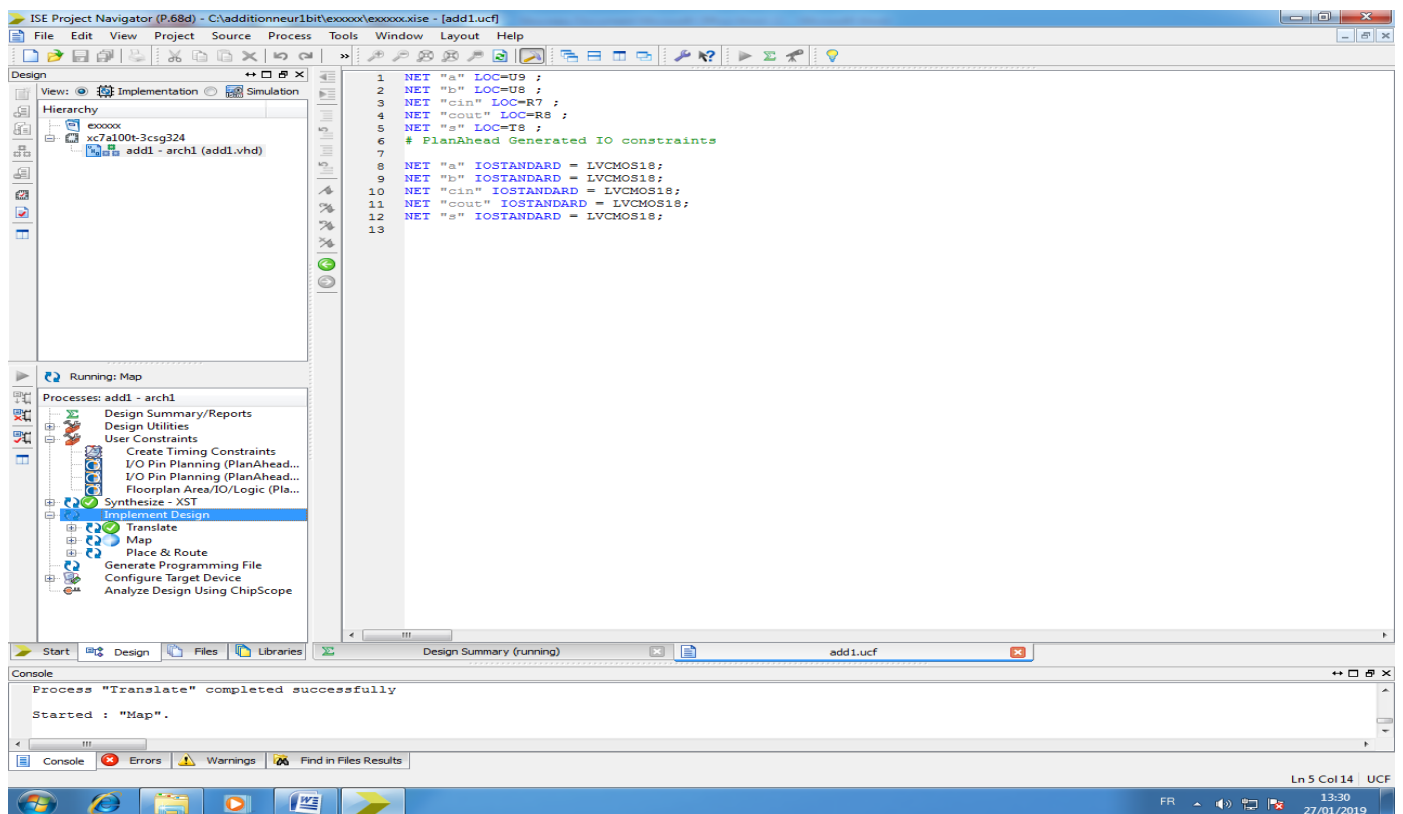
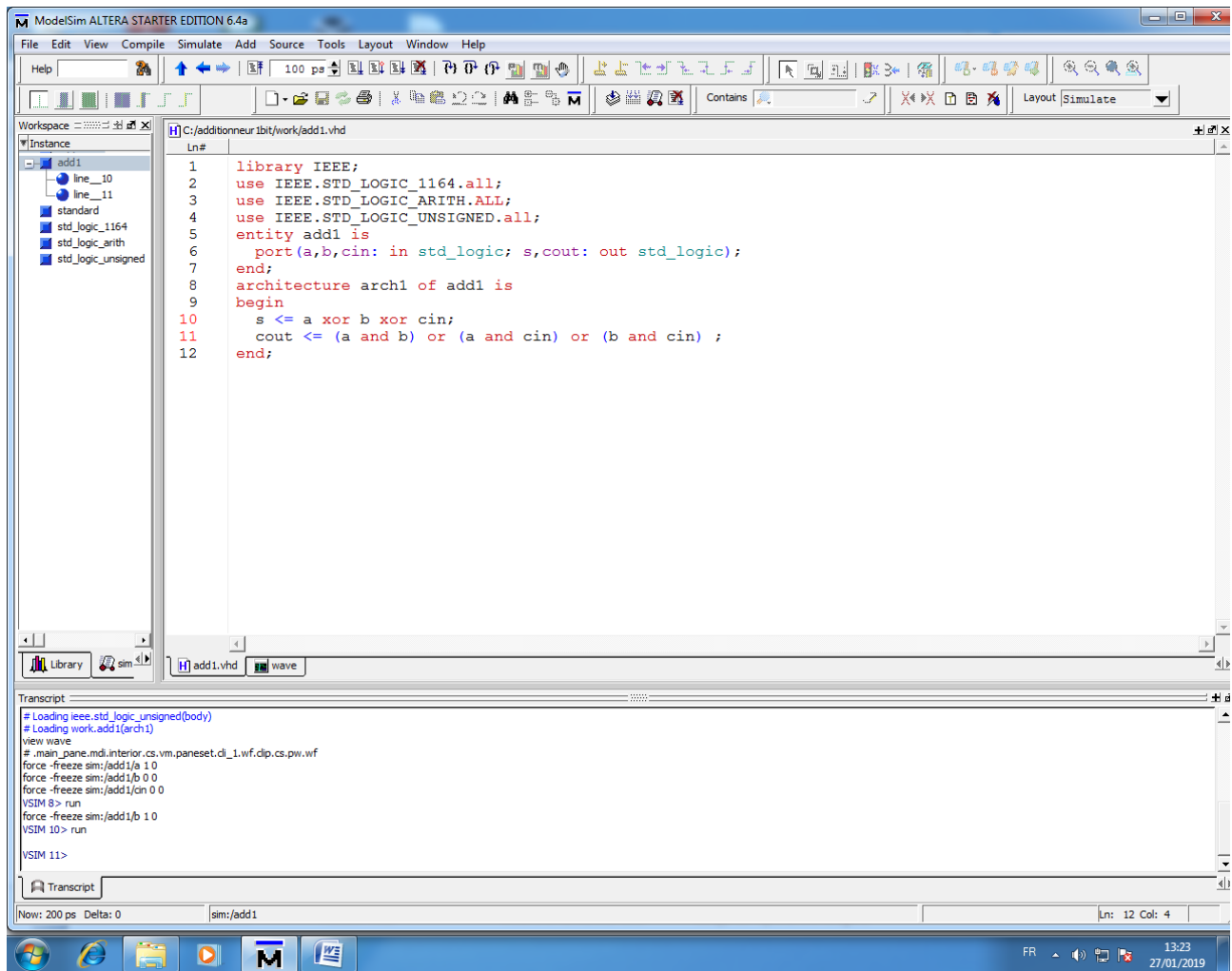
Console Errors Warnings

Configuration Digilent Nexys4 10000000

FR 13:14 27/01/2019

Exercise

- Additionneur 1 bit



ISE Project Navigator (P.68d) - C:\additionneur1bit\exxxxx\exxxxx\ise - [add1.ucf]

File Edit View Project Source Process Tools Window Layout Help

Design

View: Implementation Simulation

Hierarchy

- exxxxx
- xc7a100t-3csg324
- add1 - arch1 (add1.vhd)

No Processes Running

Processes: add1 - arch1

- Design Summary/Reports
- Design Utilities
- User Constraints
- Create Timing Constraints
- I/O Pin Planning (PlanAhead...)
- I/O Pin Planning (PlanAhead...)
- Floorplan Area/IO/Logic (Pla...
- Synthesize - XST
- Implement Design
- Translate
- Map
- Place & Route
- Generate Programming File
- Configure Target Device
- Analyze Design Using ChipScope

```
1 NET "a" LOC=U9 ;
2 NET "b" LOC=U8 ;
3 NET "cin" LOC=R7 ;
4 NET "cout" LOC=R8 ;
5 NET "s" LOC=T8 ;
6 # PlanAhead Generated IO constraints
7
8 NET "a" IOSTANDARD = LVCMOS18;
9 NET "b" IOSTANDARD = LVCMOS18;
10 NET "cin" IOSTANDARD = LVCMOS18;
11 NET "cout" IOSTANDARD = LVCMOS18;
12 NET "s" IOSTANDARD = LVCMOS18;
13
```

Start Design Files Libraries

Design Summary (Programming File Generated) add1.ucf

Console

For further information on the status of this process, see the "_impact.log" file.

Process "Configure Target Device" launched successfully

Ln 5 Col 14 UCF

13:35 27/01/2019

ISE IMPACT (P.68d) - [Boundary Scan]

File Edit View Operations Output Debug Window Help

IMPACT Flows

- Boundary Scan
- SystemACE
- Create PROM File (PROM File Format...)
- WebTalk Data

IMPACT Processes

Available Operations are:

- Program
- Get Device ID
- Get Device Signature/Usercode
- Read Device Status
- One Step SVF
- One Step XSVF

Diagram:

TDI

TDO

xc7a100t
add1.bit

Program Succeeded

Boundary Scan

Console

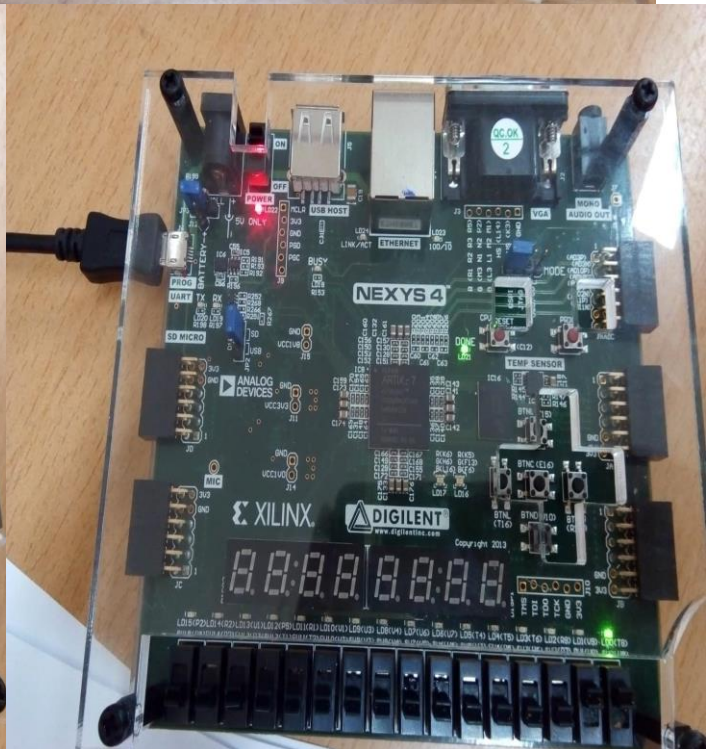
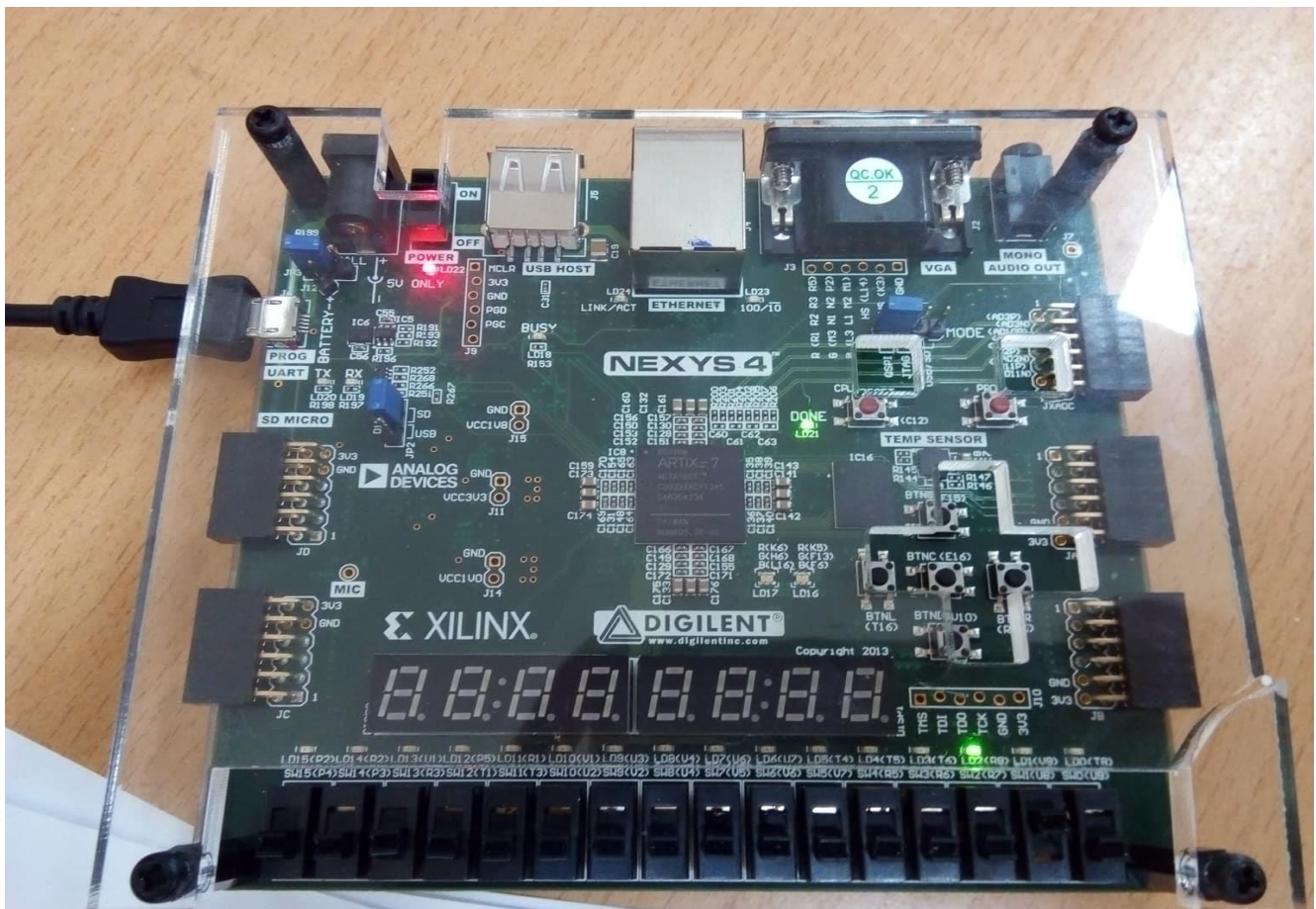
LCK cycle: NoWait

INFO:IMPACT - '1': Checking done pin....done.

'1': Programmed successfully.

Configuration | Digilent Nexys4 | 10000000

13:32 27/01/2019



Realisation d'un additionneur 4 bit :

The screenshot shows the ModelSim ALTERA STARTER EDITION 6.4a interface. The main window displays the VHDL code for a 4-bit adder. The code defines an entity 'add4' with two 4-bit inputs 'a' and 'b', a 1-bit input 'cin', and two 4-bit outputs 'cout' and 's'. The architecture 'arch1' uses a component 'add1' to implement the addition logic.

```

1  library IEEE;
2  use IEEE.STD_LOGIC_1164.all;
3  use IEEE.STD_LOGIC_ARITH.all;
4  use IEEE.STD_LOGIC_UNSIGNED.all;
5  entity add4 is
6      port(a,b: in std_logic_vector(3 downto 0); cin:in std_logic;
7  end;
8  architecture arch1 of add4 is
9      component add1
10         port(a,b,cin: in std_logic; s,cout: out std_logic);
11     end component;
12     signal n: std_logic_vector(2 downto 0);
13     begin
14         p1:add1 port map (a(0),b(0),cin,s(0),n(0));
15         p2:add1 port map (a(1),b(1),n(0),s(1),n(1));
16         p3:add1 port map (a(2),b(2),n(1),s(2),n(2));
17         p4:add1 port map (a(3),b(3),n(2),s(3),cout);
18     end arch1;
19

```

The Objects window on the left shows the signals and their values: 'a' is 1111, 'b' is 0000, 'cin' is 0, 'cout' is 0, 'n' is 000, and 's' is 1111. The Transcript window at the bottom shows the simulation setup and execution commands.

The screenshot shows the PlanAhead 14.6 interface. The main window displays the physical design of the 4-bit adder, showing the placement of logic blocks and the routing of signals. The design is organized into a grid with columns numbered 1 to 18 and rows lettered A to V. The logic blocks are represented by colored shapes, and the routing is shown as lines connecting the blocks.

The I/O Ports window at the bottom shows the configuration of the adder's ports:

Name	Direction	Neg Diff Pair	Site	Fixed	Bank	I/O Std	Vcco	Vref	Drive Stre...	Slew Type	Pull Type	Off-Chip T...	IN_TERM
a	Input		U9	☒		34 LVCMOS18	1.800				NONE	NONE	
b	Input		U8	☒		34 LVCMOS18	1.800				NONE	NONE	
cout	Output		T8	☒		34 LVCMOS18	1.800		12 SLOW		NONE	FP_VTT_50	
s	Output		R8	☒		34 LVCMOS18	1.800		12 SLOW		NONE	FP_VTT_50	

ISE Project Navigator (P.68d) - C:\48IT\bit\bit.xise - [add4.ucf]

File Edit View Project Source Process Tools Window Layout Help

Design

View: Implementation Simulation

Hierarchy

- bit
- xc7a100t-3csg324
- add4 - arch1 (add4.vhd)

Processes: add4 - arch1

- Design Summary/Reports
- Design Utilities
- User Constraints
- Create Timing Constraints
- I/O Pin Planning (PlanAhead...)
- I/O Pin Planning (PlanAhead...)
- Floorplan Area/IO/Logic (Pla...)
- Synthesize - XST
- Implement Design
- Translate
- Map
- Place & Route
- Generate Programming File
- Configure Target Device
- Generate Target PROM/ACE...
- Manage Configuration Proje...
- Analyze Design Using ChipScope

No Processes Running

Design Summary (Programming File Generated)

add4.ucf

```
1 NET "a[3]" LOC=U9 ;
2 NET "a[2]" LOC=U8 ;
3 NET "a[1]" LOC=R7 ;
4 NET "a[0]" LOC=R6 ;
5
6 NET "b[3]" LOC=R5 ;
7 NET "b[2]" LOC=V7 ;
8 NET "b[1]" LOC=V6 ;
9 NET "b[0]" LOC= V5;
10
11 NET "s[3]" LOC=T8 ;
12 NET "s[2]" LOC=V9 ;
13 NET "s[1]" LOC=R8 ;
14 NET "s[0]" LOC=T6 ;
15
16 NET "cin" LOC=U4 ;
17 NET "cout" LOC=T5 ;
18
19 NET "a[3]" IOSTANDARD = LVCMOS18;
20 NET "a[2]" IOSTANDARD = LVCMOS18;
21 NET "a[1]" IOSTANDARD = LVCMOS18;
22 NET "a[0]" IOSTANDARD = LVCMOS18;
23 NET "b[3]" IOSTANDARD = LVCMOS18;
24 NET "b[2]" IOSTANDARD = LVCMOS18;
25 NET "b[1]" IOSTANDARD = LVCMOS18;
26 NET "b[0]" IOSTANDARD = LVCMOS18;
27 NET "s[3]" IOSTANDARD = LVCMOS18;
28 NET "s[2]" IOSTANDARD = LVCMOS18;
29 NET "s[1]" IOSTANDARD = LVCMOS18;
30 NET "s[0]" IOSTANDARD = LVCMOS18;
31
32 # PlanAhead Generated IO constraints
33
34 NET "cin" IOSTANDARD = LVCMOS18;
35 NET "cout" IOSTANDARD = LVCMOS18;
36
```

Console

For further information on the status of this process, see the "_impact.log" file.

Process "Configure Target Device" launched successfully

Ln 14 Col 9 UCF

14:07 27/01/2019

ISE Project Navigator (P.68d) - C:\48IT\bit\bit.xise - [add4.ucf]

File Edit View Project Source Process Tools Window Layout Help

Design

View: Implementation Simulation

Hierarchy

- bit
- xc7a100t-3csg324
- add4 - arch1 (add4.vhd)

Processes: add4 - arch1

- Design Summary/Reports
- Design Utilities
- User Constraints
- Create Timing Constraints
- I/O Pin Planning (PlanAhead...)
- I/O Pin Planning (PlanAhead...)
- Floorplan Area/IO/Logic (Pla...)
- Synthesize - XST
- Implement Design
- Translate
- Map
- Place & Route
- Generate Programming File
- Configure Target Device
- Generate Target PROM/ACE...
- Manage Configuration Proje...
- Analyze Design Using ChipScope

Running: Generate Bitstream

Design Summary (running)

add4.ucf

```
1 NET "a[3]" LOC=U9 ;
2 NET "a[2]" LOC=U8 ;
3 NET "a[1]" LOC=R7 ;
4 NET "a[0]" LOC=R6 ;
5
6 NET "b[3]" LOC=R5 ;
7 NET "b[2]" LOC=V7 ;
8 NET "b[1]" LOC=V6 ;
9 NET "b[0]" LOC= V5;
10
11 NET "s[3]" LOC=T8 ;
12 NET "s[2]" LOC=V9 ;
13 NET "s[1]" LOC=R8 ;
14 NET "s[0]" LOC=T6 ;
15
16 NET "cin" LOC=U4 ;
17 NET "cout" LOC=T5 ;
18
19 NET "a[3]" IOSTANDARD = LVCMOS18;
20 NET "a[2]" IOSTANDARD = LVCMOS18;
21 NET "a[1]" IOSTANDARD = LVCMOS18;
22 NET "a[0]" IOSTANDARD = LVCMOS18;
23 NET "b[3]" IOSTANDARD = LVCMOS18;
24 NET "b[2]" IOSTANDARD = LVCMOS18;
25 NET "b[1]" IOSTANDARD = LVCMOS18;
26 NET "b[0]" IOSTANDARD = LVCMOS18;
27 NET "s[3]" IOSTANDARD = LVCMOS18;
28 NET "s[2]" IOSTANDARD = LVCMOS18;
29 NET "s[1]" IOSTANDARD = LVCMOS18;
30 NET "s[0]" IOSTANDARD = LVCMOS18;
31
32 # PlanAhead Generated IO constraints
33
34 NET "cin" IOSTANDARD = LVCMOS18;
35 NET "cout" IOSTANDARD = LVCMOS18;
36
```

Console

Started : "Generate Programming File".

Running bitgen...

Command Line: bitgen -intstyle ise -f add4.ut add4.mcd

Ln 14 Col 9 UCF

14:04 27/01/2019

ISE Project Navigator (P.68d) - C:\48IT\bit\bit.xise - [add4.ucf]

File Edit View Project Source Process Tools Window Layout Help

Design

View: Implementation Simulation

Hierarchy

- bit
- xc7a100t-3csg324
 - add4 - arch1 (add4.vhd)

No Processes Running

Processes: add4 - arch1

- Design Summary/Reports
- Design Utilities
- User Constraints
- Create Timing Constraints
- I/O Pin Planning (PlanAhead...)
- Floorplan Area/IO/Logic (Pla...)
- Synthesize - XST
- Implement Design
- Translate
- Map
- Place & Route
- Generate Programming File
- Configure Target Device
- Generate Target PROM/ACE ...
- Manage Configuration Proje...
- Analyze Design Using ChipScope

```
1 NET "a[3]" LOC=U9 ;
2 NET "a[2]" LOC=U8 ;
3 NET "a[1]" LOC=R7 ;
4 NET "a[0]" LOC=R6 ;
5
6 NET "b[3]" LOC=R5 ;
7 NET "b[2]" LOC=V7 ;
8 NET "b[1]" LOC=V6 ;
9 NET "b[0]" LOC=V5 ;
10
11 NET "s[3]" LOC=I8 ;
12 NET "s[2]" LOC=V9 ;
13 NET "s[1]" LOC=R8 ;
14 NET "s[0]" LOC=I6 ;
15
16 NET "cin" LOC=U4 ;
17 NET "cout" LOC=I5 ;
18
19 NET "a[3]" IOSTANDARD = LVCMOS18;
20 NET "a[2]" IOSTANDARD = LVCMOS18;
21 NET "a[1]" IOSTANDARD = LVCMOS18;
22 NET "a[0]" IOSTANDARD = LVCMOS18;
23 NET "b[3]" IOSTANDARD = LVCMOS18;
24 NET "b[2]" IOSTANDARD = LVCMOS18;
25 NET "b[1]" IOSTANDARD = LVCMOS18;
26 NET "b[0]" IOSTANDARD = LVCMOS18;
27 NET "s[3]" IOSTANDARD = LVCMOS18;
28 NET "s[2]" IOSTANDARD = LVCMOS18;
29 NET "s[1]" IOSTANDARD = LVCMOS18;
30 NET "s[0]" IOSTANDARD = LVCMOS18;
31
32 # PlanAhead Generated IO constraints
33
34 NET "cin" IOSTANDARD = LVCMOS18;
35 NET "cout" IOSTANDARD = LVCMOS18;
36
```

Start Design Files Libraries

Design Summary (Programming File Generated) add4.ucf

Console

For further information on the status of this process, see the "_impact.log" file.

Process "Configure Target Device" launched successfully

Ln 14 Col 9 UCF

14:07 27/01/2019

ISE iMPACT (P.68d) - [Boundary Scan]

File Edit View Operations Output Debug Window Help

IMPACT Flows

- Boundary Scan
- SystemACE
- Create PROM File (PROM File Format...)
- Web Talk Data

IMPACT Processes

Available Operations are:

Device Programming Properties - Device 1 Programming Properties

Category

- Boundary-Scan
 - Device 1 (FPGA xc7a100t)

Property Name Value

Verify ☐

OK Cancel Apply Help

Identify Succeeded

Boundary Scan

Console

INFO:IMPACT:501 - '1': Added Device xc7a100t successfully.

Configuration Digilent Nexys4 10000000

13:15 27/01/2019

ISE iMPACT (P.68d) - [Boundary Scan]

File Edit View Operations Output Debug Window Help

IMPACT Flows

- Boundary Scan
- SystemACE
- Create PROM File (PROM File Format...)
- WebTalk Data

IMPACT Processes

Configuration Operation Status

Executing command...

33%

Abort

Console

```
INFO:iMPACT - Creating XC7A100T device.  
PROGRESS START - Starting Operation.  
'1': Programming device...
```

Configuration | Digilent Nexys4 | 10000000

FR 13:15 27/01/2019

ISE iMPACT (P.68d) - [Boundary Scan]

File Edit View Operations Output Debug Window Help

IMPACT Flows

- Boundary Scan
- SystemACE
- Create PROM File (PROM File Format...)
- WebTalk Data

IMPACT Processes

Available Operations are:

- Program
- Get Device ID
- Get Device Signature/Usercode
- Read Device Status
- One Step SVF
- One Step XSUF

Program Succeeded

Console

```
LCK cycle: NoWait  
INFO:iMPACT - '1': Checking done pin....done.  
'1': Programmed successfully.
```

Configuration | Digilent Nexys4 | 10000000

FR 13:16 27/01/2019

